Speaker Information

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Talk Information

How ocean acidification rewires squid optic lobes and breaks hunting

Bigfin reef squid (*Sepioteuthis lessoniana*) are agile predators whose success depends on rapid visual processing and precisely timed strikes. We asked what happens to that system under chronic ocean acidification. Laboratory-reared squid exposed from hatching to pH 7.8, the projected year-2100 condition, showed substantially reduced predatory success after 90 days. The retina still worked. Custom electroretinogram recordings confirmed that peripheral photoreception remained functional across light intensities, prompting attention to what lies beyond the eye: the optic lobes. There, acetylcholine receptor expression was down, systemic HCO_3^- was elevated, and metabolic rates had shifted. Transcriptomics fleshed out the timeline. Acute exposure suppressed energy metabolism and synaptic signaling broadly; chronic exposure brought partial compensation through upregulated cellular maintenance. The squid adapted, but not enough to hunt normally. Acid-base disruption, cholinergic signaling, and behavioral failure appear connected at the level of neural integration, not at the sensory periphery. What that means for predator-prey dynamics in a high-CO_2 ocean is something we are only beginning to measure.

Research Overview (for collaboration discussion)

[Research Interests]
Comparative marine animal physiology, with a focus on acid-base regulation, neural and behavioral responses to environmental stress, and adaptation mechanisms under ocean acidification and hydrothermal vent conditions
[Current Research Projects]
(1) Neuro-behavioral physiology of cephalopods under ocean acidification; (2) Extreme environment adaptation in the hydrothermal vent crab, including host-symbiont interactions; (3) Multi-generational epigenetic mechanisms of CO_2 adaptation in fish
[Techniques]
Whole-animal behavioral assays, electroretinography, transcriptomics (RNA-seq, scRNA-seq), acid-base physiology (blood gas analysis, ion transport measurements), SCUBA-based field sampling
[Aspects or techniques of interest for future collaboration]
In vivo neural imaging and electrophysiology approaches applicable to cephalopod central nervous systems; methods for dissecting sensory processing versus higher-order integration; comparative neurophysiology frameworks linking environmental physiology with behavioral outcomes

Short Bio

Yung-Che Tseng is an Associate Research Fellow at Academia Sinica's Marine Research Station (ICOB), Taiwan. His research examines how marine animals respond physiologically and neurally to environmental change as well as extreme marine settings, with a particular focus on cephalopods, hydrothermal vent organisms, and multi-generational epigenetic adaptation.